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Thermodynamic assessment of the Co–Pr, Er–Ni and Ni–Pr systems

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Abstract

By means of CALPHAD technique, the Co–Pr, Er–Ni and Ni–Pr systems were critically assessed. The solution phases (liquid, bcc, fcc, hcp and dhcp) are modeled with the Redlich–Kister equation, and the intermetallic compounds treated as stoichiometric compounds. Self-consistent thermodynamic parameters of the Co–Pr, Er–Ni and Ni–Pr systems were obtained. The calculations agree well with experiments. © 1999 Elsevier Science S.A. All rights reserved.

Keywords: Thermodynamic properties; Co–Pr system; Er–Ni system; Ni–Pr system

1. Introduction

The intermetallic compounds formed by rare earth elements and transition metals are of particular interest regarding their outstanding magnetic properties [1] and their reversible absorption of hydrogen gas at room temperature and nearly atmospheric pressure [2]. It is well known that the phase diagrams and thermodynamic properties of the alloy system are of great value to alloy design and processing.

This work deals with the assessment of the thermodynamic descriptions of the Co–Pr, Er–Ni and Ni–Pr systems by means of the CALPHAD technique. In this method, the thermodynamic properties of the alloy systems are studied by using thermodynamic models for the Gibbs energy of individual phases. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamic and phase diagram information.

2. Thermodynamic models

The Gibbs energy of individual phases is described by the sublattice model [3] relative to the Standard Element Reference (SER), i.e. the enthalpies of the pure elements in their defined reference phase at 298.15 K.

2.1. Liquid, bcc, fcc, hcp and dhcp phases

The Gibbs energies of the liquid, bcc, fcc, hcp and dhcp phases are described by the following expression:

$$G_m^\phi = x_A^0 G_A^\phi + x_B^0 G_B^\phi + RT(x_A \ln x_A + x_B \ln x_B) + {}^E G_m^\phi + {}^{mg} G_m^\phi \quad (1)$$

where ${}^0G_i^\phi$ is the molar Gibbs energy of the element i (i = A,B respectively) with the structure ϕ in a nonmagnetic state from the compilation by Dinsdale [4]; ${}^E G_m^\phi$ is the excess Gibbs energy, expressed in Redlich–Kister polynomials,

$${}^E G_m^\phi = x_A x_B \sum_i {}^i L^\phi (x_A - x_B)^i \quad (2)$$

where ${}^iL^\phi$ is the interaction parameter between elements A and B, which is to be evaluated in present work. It's general form is

$$L^\phi = a + bT + cT \ln T + dT^2 + eT^3 + fT^{-1} \quad (3)$$

but there are cases when only the first one or two terms are used. ${}^{mg}G_m^\phi$ is the magnetic contribution to Gibbs energy in solution phases, and expressed as following [5]:

$${}^{mg} G_m^\phi = RT \ln(\beta^\phi + 1) f(\tau^\phi) \quad (4)$$

where β^ϕ is a quantity related to the total magnetic entropy which in most cases is set equal to the Bohr magnetic moment per mole of atoms; τ^ϕ is defined as T/T_c^ϕ, and T_c^ϕ is the critical temperature for magnetic

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ordering, i.e. the Curie temperature (T_c) for ferromagnetic ordering and the Neel temperature (T_N) for antiferromagnetic ordering. They are described by the following expressions:

$$T_c^{\phi} = x_A^0 T_{cA}^{\phi} + x_B^0 T_{cB}^{\phi} + x_A x_B L_{Tc}^{\phi} \quad (5)$$

$$\beta^{\phi} = x_A^0 \beta_A^{\phi} + x_B^0 \beta_B^{\phi} + x_A x_B L_{\beta}^{\phi} \quad (6)$$

where L_{Tc}^{ϕ} and L_{β}^{ϕ} are the magnetic interaction parameters between the elements A and B, here set to be zero due to lack of experimental data and extremely low solubilities in three binaries. The f(τ) represents the polynomials obtained by Hillert and Jarl [5] based on the magnetic specific heat of iron, i.e. for $\tau<1$:

$$f(\tau) = 1 - [79\tau^{-1}/(140p) + 474/497(1/p-1)(\tau^3/6 + \tau^9/135 + \tau^{15}/600)]/A \quad (7)$$

for $\tau>1$:

$$f(\tau) = -(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500)/A \quad (8)$$

where $A=519/1125+11692/15975(1/p-1)$ and p depends on the structure, 0.4 for bcc structure and 0.28 for the others.

2.2. Intermetallic compounds

All intermetallic compounds are treated by a two-sublattice model with A on one sublattice and B on the other one. Owing to lack of experimental measurements, it is assumed that the Neumann–Kopp rule applies to the heat capacity, i.e. $\Delta C_p=0$. Thus the Gibbs energy for per mole of formula unit A_mB_n can be expressed as following:

$$G_m^{A_mB_n} = m^0 G_A^{ref} + n^0 G_B^{ref} + a + bT + {}^{mg}G_m^{A_mB_n} \quad (9)$$

where ${}^0G_i^{ref}$ is the molar Gibbs energy of the pure element i (i=A,B respectively) with its defined reference structure in a nonmagnetic state from the compilation by Dinsdale [4]; The parameters a and b were evaluated in the present work. The magnetic contribution to the Gibbs energy ${}^{mg}G_m^{AmBn}$ is similarly given by Eq. (4), Eq. (7) and Eq. (8), the T_c and β are taken from the measured critical temperature and the mean Bohr magnetic moment per mole of formula unit.

3. Experimental information

3.1. Co–Pr system

The complete phase diagram of the Co–Pr system was first measured by Ray et al. [6–8]. Nine intermetallic compounds were reported: $Co_{17}Pr_2$, Co_5Pr, $Co_{19}Pr_5$, Co_7Pr_2, Co_3Pr, Co_2Pr, $Co_{1.7}Pr_2$, $Co_{\sim3}Pr_{\sim7}$ and $CoPr_3$. Except the congruently melting $CoPr_3$ compound, these compounds form peritectically. Massalski et al. [9] modified the compound $Co_{\sim3}Pr_{\sim7}$ to Co_2Pr_5 based on the crystal structure data given by Pearson [10].

More recently the Co–Pr system was revestigated by Wu et al. [11], who observed eight intermetallic compounds and specified the compound formerly called $Co_{1.7}Pr_2$ to be Co_3Pr_4. The compound Co_5Pr_2 was not observed. The results are mostly in good agreement with results of Ray [8] except for the type of melting of $Co_{17}Pr_2$ and the presence or absence of Co_5Pr_2, as discussed by Okamoto [12]. Wu et al. [11] suggested that the compound $Co_{17}Pr_2$ formed congruently with a eutectic reaction liquid→$Co_{17}Pr_2$+ fcc(Co) at 93.0 at.% Co, but the microstructure of the as-cast alloy containing 92.9 at.%Co given by Wu et al. [11] does not exhibit the characteristics of a eutectic structure. The suggestion by Ray [8] that $Co_{17}Pr_2$ is a peritectical compound is more likely. The solid solubility of Pr in Co at 1000°C given by Wu et al. [11] is extremely small (~0.03 at.%) and therefore is neglected in the present work.

Chuang et al. [13] and Buschow [14] reported that Co_5Pr is unstable at low temperature, and decomposes eutectoidally into $Co_{17}Pr_2$ and $Co_{19}Pr_5$. The enthalpies of formation of the Co_2Pr was measured by Deodhar et al. [15]. Buschow [1] summarized the magnetic data of the compounds which are reproduced in Table 1. Buschow [16], Schweizer [17] and Khan [18] also studied on the Co–Pr system.

3.2. Er–Ni system

The phase diagram of the Er–Ni system was first measured by Buschow [19], and eleven intermetallic compounds were identified: Er_3Ni, Er_5Ni_3, ErNi, $ErNi_2$, $ErNi_3$, Er_2Ni_7, $ErNi_4$, Er_4Ni_{17}, Er_5Ni_{22}, $ErNi_5$ and Er_2Ni_{17}. Two compounds ErNi and $ErNi_5$ melt congruently, whereas the others are peritectic or peritectoidic compounds. $ErNi_5$ was observed to have a small homogeneity region on the Ni-rich side at high temperature, but in the present work it is treated as stoichiometric com-

Table 1
Measured T_c and $\beta(\mu_B)$ values in the Co–Pr system

Phase	$Co_{17}Pr_2$	Co_5Pr	$Co_{19}Pr_5$	Co_7Pr_2	Co_3Pr	Co_2Pr
T_c(K)	1177	912	690	574	339	49
$\beta(\mu_B)$	32	9.95	2.97	10.5	3.8	2.83

Table 2
Measured T_c and $\beta(\mu_B)$ values in the Er–Ni system

Phase	Er_3Ni	ErNi	$ErNi_2$	$ErNi_3$	Er_2Ni_7	$ErNi_5$	Er_2Ni_{17}
T_c(K)	7.7	12	18.5	64	74	12	166
$\beta(\mu_B)$	7.1	8.1	6.9	6.08	5.82	7.7	10

pound because of limited information. The solubilities of Ni in hcp(Er) is negligible and no homogeneity region of fcc(Ni) observed.

The investigation given by Moreau et al. [20] indicated that Er_5Ni_3 should be replaced by Er_3Ni_2. The enthalpies of formation of the Er_3Ni, ErNi, $ErNi_2$, $ErNi_5$ and Er_2Ni_{17} were measured by Deodhar et al. [15] and Schott et al. [21]. The magnetic properties of the Er–Ni compounds were investigated systematically by Buschow [19] and Abrahams et al. [22], and are presented in Table 2. Because the Curie temperature of all compounds in the Er–Ni system is below 298.15 K, the magnetic contribution to the Gibbs energy is in the non-magnetic term. Recently the Er–Ni system was reviewed by Pan and Nash [23].

3.3. Ni–Pr system

The Ni–Pr system was first studied by Vogel [24]. Six compounds were reported: $NiPr_3$, NiPr, Ni_2Pr, Ni_3Pr, Ni_4Pr and Ni_5Pr, but the existence of Ni_3Pr and Ni_4Pr was not certain. Lemaire et al. [25] confirmed the existence of Ni_3Pr. Gschneidner [26] proposed that the true composition of Ni_4Pr should be Ni_7Pr_2, which was confirmed by Buschow et al. [27] and Taylor [28]. The occurrence of Ni_3Pr_7 was ascertained by Kissel et al. [29] and Olcese [30]. Vogel [24] suggested that Ni_2Pr decomposes peritectically at 880°C, but Wernick et al. [31] did not observe any transition between room temperature and 880°C. Qi et al. [32] measured the liquidus in equilibrium with the Ni_5Pr compound.

Recently Pan and Cheng [33] revestigated the Ni–Pr system. All seven previously reported compounds were comfirmed. The constructed phase diagram is similar to the evaluation given by Gschneidner [26], but the temperature and compositions of invariant reactions are rather different. The results of Pan and Cheng [33] are accepted in the present work, because they used higher purity starting metals than Vogel [24]. The Ni–Pr system was reviewed by Pan and Nash [34].

The enthalpies of formation of Ni_2Pr and Ni_5Pr were measured by Deodhar et al. [15] and Pasturel et al. [35], respectively. The magnetic properties of the Ni–Pr compounds were measured by Abrahams et al. [22], Walline et al. [36] and Lemaire et al. [37]. Because the Curie temperature of all compounds in the Er–Ni system is much below 298.15 K, the magnetic contribution to the Gibbs energy is in the non–magnetic term.

4. Assessment procedure

Most of the experimental information mentioned above has been selected for the evaluation of the thermodynamic model parameters. For magnetic properties, the T_c and β values are simply taken from the experimental measurements.

In the Co–Pr system, there are five solution phases: liquid, bcc, fcc, hcp and dhcp. As the dhcp structure is not stable for pure Co and the hcp structure is not stable for pure Pr, their lattice stabilities are required. According to the estimation of Fernandez Guillermet and Huang [38] for high melting bcc metals V, Nb and Ta, the following lattice stabilities are estimated,

$$^0G_{Co}^{dhcp} = {}^0G_{Co}^{hcp} + 5000 \text{ J mol}^{-1} \tag{10}$$

$$^0G_{Pr}^{hcp} = {}^0G_{Pr}^{dhcp} + 5000 \text{ J mol}^{-1} \tag{11}$$

No solubility is assumed in the bcc, fcc, hcp and dhcp phases, which is realized by assigning a large positive interaction parameter, i.e.

$$^0L_{Co,Pr}^{bcc} = {}^0L_{Co,Pr}^{fcc} = {}^0L_{Co,Pr}^{hcp} = {}^0L_{Co,Pr}^{dhcp} = 10^5 \text{ J mol}^{-1} \tag{12}$$

In the Er–Ni system, the following lattice stabilities are estimated based on same reason discussed in the Co–Pr system above,

$$^0G_{Er}^{fcc} = {}^0G_{Er}^{hcp} + 5000 \text{ J mol}^{-1} \tag{13}$$

$$^0G_{Ni}^{hcp} = {}^0G_{Ni}^{fcc} + 5000 \text{ J mol}^{-1} \tag{14}$$

Table 3
Optimized thermodynamic parameters of the Co–Pr system, in SI units

Phase	Parameters
Liquid	$^0L^{liquid} = -68784 + 31.596T$
	$^1L^{liquid} = -17505 + 8.223T$
bcc	$^0L^{bcc} = 100000$
fcc	$^0L^{fcc} = 100000$
hcp	$^0L^{hcp} = 100000$
dhcp	$^0L^{dhcp} = 100000$
$Co_{17}Pr_2$	$G_m^{Co_{17}Pr_2} - 17\,^0G_{Co}^{hcp} - -2\,^0G_{Pr}^{dhcp} = -128412 + 12.793T$
Co_5Pr	$G_m^{Co_5Pr} - 5\,^0G_{Co}^{hcp} - {}^0G_{Pr}^{dhcp} = -63311 + 9.468T$
$Co_{19}Pr_5$	$G_m^{Co_{19}Pr_5} - 19\,^0G_{Co}^{hcp} - 5\,^0G_{Pr}^{dhcp} = -386270 + 115.310T$
Co_7Pr_2	$G_m^{Co_7Pr_2} - 7\,^0G_{Co}^{hcp} - 2\,^0G_{Pr}^{dhcp} = -144708 + 41.081T$
Co_3Pr	$G_m^{Co_3Pr} - 3\,^0G_{Co}^{hcp} - {}^0G_{Pr}^{dhcp} = -64555 + 17.009T$
Co_2Pr	$G_m^{Co_2Pr} - 2\,^0G_{Co}^{hcp} - {}^0G_{Pr}^{dhcp} = -45681 + 8.915T$
Co_2Pr_5	$G_m^{Co_2Pr_5} - 2\,^0G_{Co}^{hcp} - 5\,^0G_{Pr}^{dhcp} = -53050 - 1.879T$
Co_3Pr_4	$G_m^{Co_3Pr_4} - 3\,^0G_{Co}^{hcp} - 4\,^0G_{Pr}^{dhcp} = -73519 + 2.375T$
$CoPr_3$	$G_m^{CoPr_3} - {}^0G_{Co}^{hcp} - 3\,^0G_{Pr}^{dhcp} = -27360 - 1.441T$

Table 4
Optimized thermodynamic parameters of the Er–Ni system, in SI units

Phase	Parameters
Liquid	${}^0L^{\text{liquid}} = -164148 + 47.642T$
	${}^1L^{\text{liquid}} = 82460 - 51.370T$
fcc	${}^0L^{\text{fcc}} = 100000$
hcp	${}^0L^{\text{hcp}} = 100000$
Er_3Ni	$G_m^{Er_3Ni} - 3{}^0G_{Er}^{hcp} - {}^0G_{Ni}^{fcc} = -66049 + 0.511T$
Er_3Ni_2	$G_m^{Er_3Ni_2} - 3{}^0G_{Er}^{hcp} - 2{}^0G_{Ni}^{fcc} = -132207 + 9.427T$
ErNi	$G_m^{ErNi} - {}^0G_{Er}^{hcp} - {}^0G_{Ni}^{fcc} = -61285 + 3.321T$
$ErNi_2$	$G_m^{ErNi_2} - {}^0G_{Er}^{hcp} - 2{}^0G_{Ni}^{fcc} = -110165 + 20.413T$
$ErNi_3$	$G_m^{ErNi_3} - {}^0G_{Er}^{hcp} - 3{}^0G_{Ni}^{fcc} = -158553 + 43.055T$
Er_2Ni_7	$G_m^{Er_2Ni_7} - 2{}^0G_{Er}^{hcp} - 7{}^0G_{Ni}^{fcc} = -325191 + 85.759T$
$ErNi_4$	$G_m^{ErNi_4} - {}^0G_{Er}^{hcp} - 4{}^0G_{Ni}^{fcc} = -163912 + 40.943T$
Er_4Ni_{17}	$G_m^{Er_4Ni_{17}} - 4{}^0G_{Er}^{hcp} - 17{}^0G_{Ni}^{fcc} = -657453 + 160.817T$
Er_5Ni_{22}	$G_m^{Er_5Ni_{22}} - 5{}^0G_{Er}^{hcp} - 22{}^0G_{Ni}^{fcc} = -823178 + 199.197T$
$ErNi_5$	$G_m^{ErNi_5} - {}^0G_{Er}^{hcp} - 5{}^0G_{Ni}^{fcc} = -165724 + 38.720T$
Er_2Ni_{17}	$G_m^{Er_2Ni_{17}} - 2{}^0G_{Er}^{hcp} - 17{}^0G_{Ni}^{fcc} = -341853 + 79.625T$

$$ {}^0L_{\text{Er,Ni}}^{\text{fcc}} = {}^0L_{\text{Er,Ni}}^{\text{hcp}} = 10^5 \text{ J mol}^{-1} \tag{15} $$

In the Ni–Pr system,

$$ {}^0G_{\text{Ni}}^{\text{dhcp}} = {}^0G_{\text{Ni}}^{\text{fcc}} + 5000 \text{ J mol}^{-1} \tag{16} $$

$$ {}^0L_{\text{Ni,Pr}} = {}^0L_{\text{Ni,Pr}}^{\text{fcc}} = {}^0L_{\text{Ni,Pr}}^{\text{dhcp}} = 10^5 \text{ J mol}^{-1} \tag{17} $$

The optimization was carried out by means of the Thermo–Calc software [39] which can handle various kinds of experimental data. The program works by minimizing an error sum between experimental and computed values. In the optimization procedure, each of the selected experimental data is given a certain weight. The weight is chosen by personal judgment and changed by trial and error during the work until most of the selected experimental information is reproduced within the expected uncertainty limits.

Table 5
Optimized thermodynamic parameters of the Ni–Pr system, in SI units

Phase	Parameters
liquid	${}^0L^{\text{liquid}} = -228244 + 157.001T$
	${}^1L^{\text{liquid}} = -40614 + 8.343T$
fcc	${}^0L^{\text{fcc}} = 100000$
bcc	${}^0L^{\text{bcc}} = 100000$
dhcp	${}^0L^{\text{dhcp}} = 100000$
Ni_5Pr	$G_m^{Ni_5Pr} - 5{}^0G_{Ni}^{fcc} - {}^0G_{Pr}^{dhcp} = -179191 + 92.421T$
Ni_7Pr_2	$G_m^{Ni_7Pr_2} - 7{}^0G_{Ni}^{fcc} - 2{}^0G_{Pr}^{dhcp} = -313084 + 166.345T$
Ni_3Pr	$G_m^{Ni_3Pr} - 3{}^0G_{Ni}^{fcc} - {}^0G_{Pr}^{dhcp} = -148365 + 80.071T$
Ni_2Pr	$G_m^{Ni_2Pr} - 2{}^0G_{Ni}^{fcc} - {}^0G_{Pr}^{dhcp} = -131998 + 74.697T$
NiPr	$G_m^{NiPr} - {}^0G_{Ni}^{fcc} - {}^0G_{Pr}^{dhcp} = -115458 + 76.530T$
Ni_3Pr_7	$G_m^{Ni_3Pr_7} - 3{}^0G_{Ni}^{fcc} - 7{}^0G_{Pr}^{dhcp} = -353601 + 215.274T$
$NiPr_3$	$G_m^{NiPr_3} - {}^0G_{Ni}^{fcc} - 3{}^0G_{Pr}^{dhcp} = -119501 + 72.087T$

5. Results and discussion

The thermodynamic descriptions of the Co–Pr, Er–Ni and Ni–Pr systems obtained in present work are listed in Tables 3–5, respectively.

The Co–Pr, Er–Ni and Ni–Pr phase diagrams calculated by means of the thermodynamic parameters are presented in Figs. 1–3 with experimental data [7,19,24,32,33], respectively. The invariant equilibria in the three systems are listed in Tables 6–8. As shown in the tables, satisfactory agreement is obtained between the presently calcula-

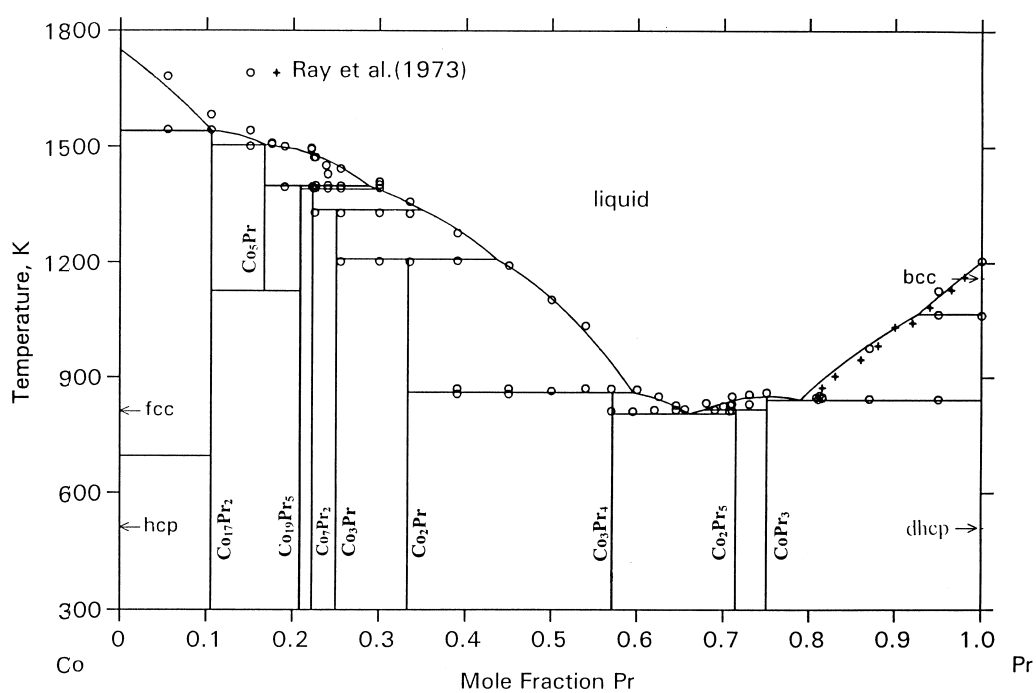

Fig. 1. The Co–Pr phase diagram calculated by present thermodynamic description with experimental measurements [7].

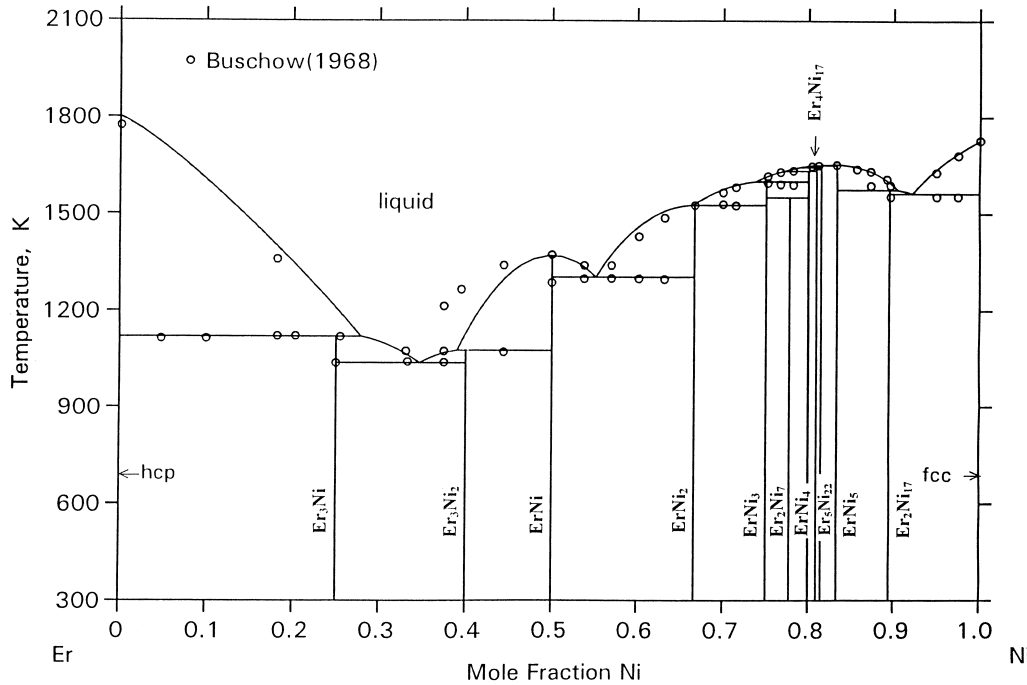

Fig. 2. The Er–Ni phase diagram calculated by present thermodynamic description with experimental measurements [19].

tions and experiments, except for the invariant temperatures of the two reactions liquid→Er_2Ni_{17}+fcc(Ni) in the Er–Ni system and liquid→$NiPr_3$+ Ni_3Pr_7 in the Ni–Pr system give large uncertainties. In view of the estimated experimental errors (about 1 to 2 at.%), twenty-six of the twenty-nine experimental invariant reaction compositions in the three systems are well reproduced.

The enthalpies of formation of the intermetallic compounds in the Co–Pr, Er–Ni and Ni–Pr systems calculated by means of the optimized thermodynamic parameters is shown in Figs. 4–6 together with the experimental data [15,21,35], respectively. As can be seen, the agreement with the data of Schott et al. [21] and Pasturel et al. [35] is good, not very good with the data of Deodhar et al. [15].

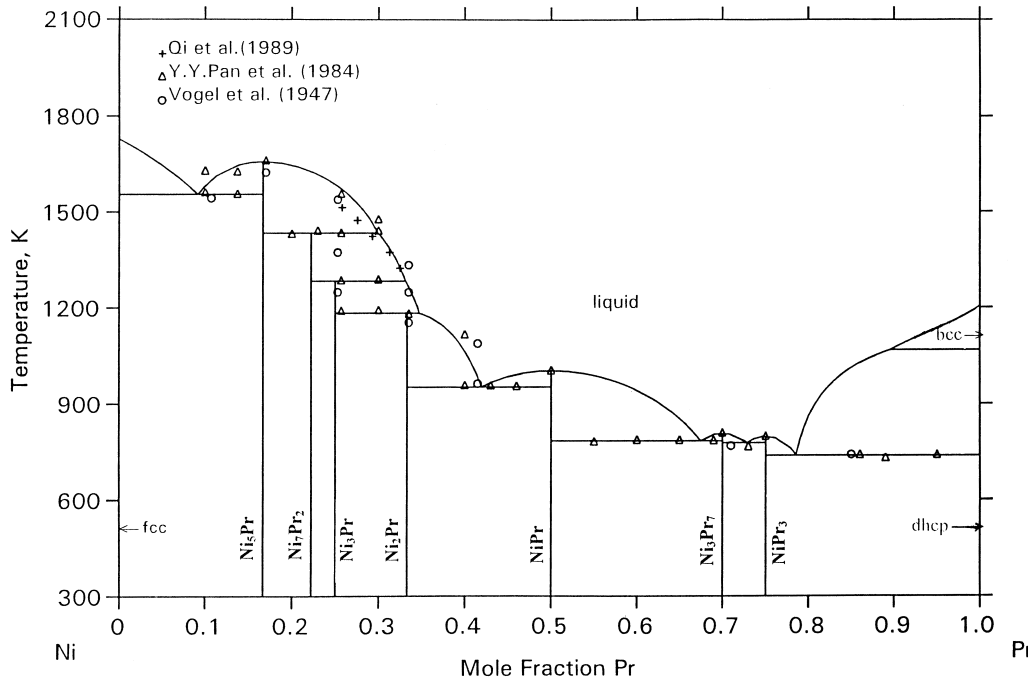

Fig. 3. The Ni–Pr phase diagram calculated by present thermodynamic description with experimental measurements [24,32,33].

Table 6
Invariant reactions in the Co–Pr system

Reactions	Present work Temperature, K (x_{Pr}^{liq})		
		[7]	[11]
liq. + fcc(Co) → $Co_{17}Pr_2$	1541(0.1059)	1544(0.145)	–
liq. + $Co_{17}Pr_2$ → Co_5Pr	1505(0.1677)	1502(0.17)	1504(0.221)
liq. + Co_5Pr → $Co_{19}Pr_5$	1400(0.2881)	1398(0.305)	1403(0.295)
Co_5Pr → $Co_{17}Pr_2$ + $Co_{19}Pr_5$	1127(–)	973(–) [14][a]	1123(–) [13]
liq. + $Co_{19}Pr_5$ → Co_7Pr_2	1391(0.2978)	1391(0.31)	1393(0.31)
liq. + Co_7Pr_2 → Co_3Pr	1338(0.3492)	1330(0.34)	1338(0.345)
liq. + Co_3Pr → Co_2Pr	1210(0.4355)	1204(0.44)	1211(0.44)
liq. + Co_2Pr → Co_3Pr_4	861(0.5950)	871(0.60)[b]	861(0.595)
liq. → Co_3Pr_4 + Co_2Pr_5	806(0.6607)	814(0.66)[b,c]	806(0.66)[d]
liq. + $CoPr_3$ → Co_2Pr_5	817(0.6776)	831(0.68)[c]	–
liq. → $CoPr_3$	852(0.75)	863(0.75)	852(0.75)
liq. → $CoPr_3$ + dhcp(Pr)	842(0.7890)	843(0.805)	843(0.805)

[a] [14] reported that Co_5Pr decomposes eutectoidally into $Co_{17}Pr_2$ and Co_7Pr_2.
[b] Co_3Pr_4 is instead of $Co_{1.7}Pr_2$.
[c] Co_2Pr_5 is instead of Co_3Pr_7.
[d] Co_2Pr_5 is instead of $CoPr_3$.

Table 7
Invariant reactions in the Er–Ni system

Reaction	Temperature, K (x_{Ni}^{liq})	
	Present work	[19]
liq. + hcp(Er) → Er_3Ni	1110(0.2770)	1118(0.26)
liq. → Er_3Ni + Er_3Ni_2	1040(0.3420)	1038(0.31)[a]
liq. + ErNi → Er_3Ni_2	1076(0.3810)	1073(0.32)[a]
liq. → ErNi	1363(0.50)	1373(0.50)
liq. → ErNi + $ErNi_2$	1306(0.5517)	1298(0.55)
liq. + $ErNi_3$ → $ErNi_2$	1527(0.6700)	1528(0.66)
liq. + $ErNi_4$ → $ErNi_3$	1595(0.7395)	1593(0.72)
$ErNi_3$ + $ErNi_4$ → Er_2Ni_7	1548(–)	1548(–)
liq. + Er_4Ni_{17} → $ErNi_4$	1634(0.7768)	1633(0.77)
liq. + Er_5Ni_{22} → Er_4Ni_{17}	1644(0.7930)	1643(0.795)
liq. + $ErNi_5$ → Er_5Ni_{22}	1651(0.8125)	1648(0.81)
liq. → $ErNi_5$	1657(0.8333)	1653(0.83)
liq. + $ErNi_5$ → Er_2Ni_{17}	1587(0.8972)	1588(0.905)
liq. → Er_2Ni_{17} + fcc(Ni)	1570(0.9240)	1553(0.92)

[a] Er_3Ni_2 is instead of Er_5Ni_3.

Table 8
Invariant reations in the Ni–Pr system

Reaction	Temperature, K (x_{Ni}^{liq})	
	Present work	[34]
liq. → Ni_5Pr + fcc(Ni)	1555(0.9083)	1553(0.89)
liq. → Ni_5Pr	1657(0.8333)	1658(0.83)
liq. + Ni_5Pr → Ni_7Pr_2	1434(0.7006)	1433(0.71)
liq. + Ni_7Pr_2 → Ni_3Pr	1283(0.6687)	1283(0.65)
liq. + Ni_3Pr → Ni_2Pr	1183(0.6080)	1183(0.62)
liq. → NiPr + Ni_2Pr	952(0.5806)	953(0.53)
liq. → NiPr	1003(0.50)	1003(0.50)
liq. → NiPr + Ni_3Pr_7	783(0.3249)	783(0.325)
liq. → Ni_3Pr_7	807(0.30)	808(0.30)
liq. → $NiPr_3$ + Ni_3Pr_7	777(0.2712)	763(0.27)
liq. → $NiPr_3$	797(0.25)	798(0.25)
liq. → dhcp(Pr) + $NiPr_3$	738(0.2144)	733(0.19)

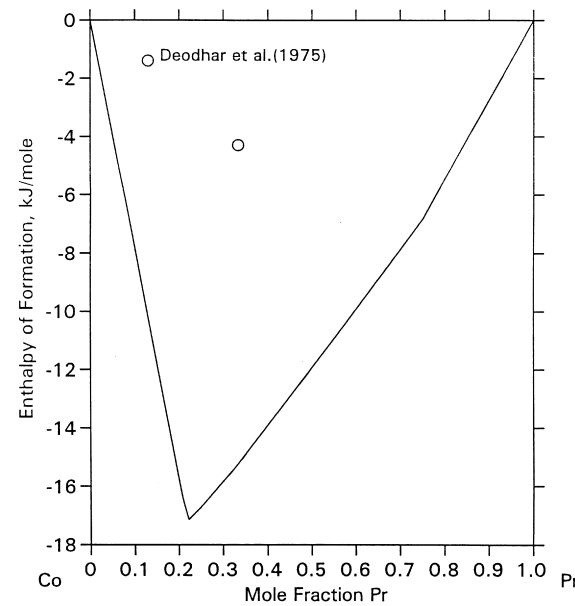

Fig. 4. Calculated enthalpy of formation in the Co–Pr system with experimental measurements [15].

Possibly the results of Deodhar et al. [15] are not very precise.

6. Conclusions

The phase relations and thermodynamic descriptions in the Co–Pr, Er–Ni and Ni–Pr systems were evaluated from the experimental information available in the literature. Consistent sets of thermodynamic parameters were derived. With the thermodynamic description available, one can now make various calculations of practical interest.

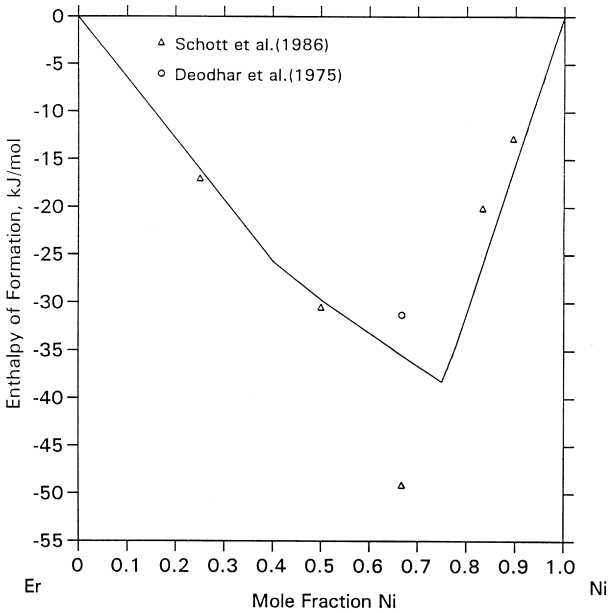

Fig. 5. Calculated enthalpy of formation in the Er–Pi system with experimental measurements [15,21].

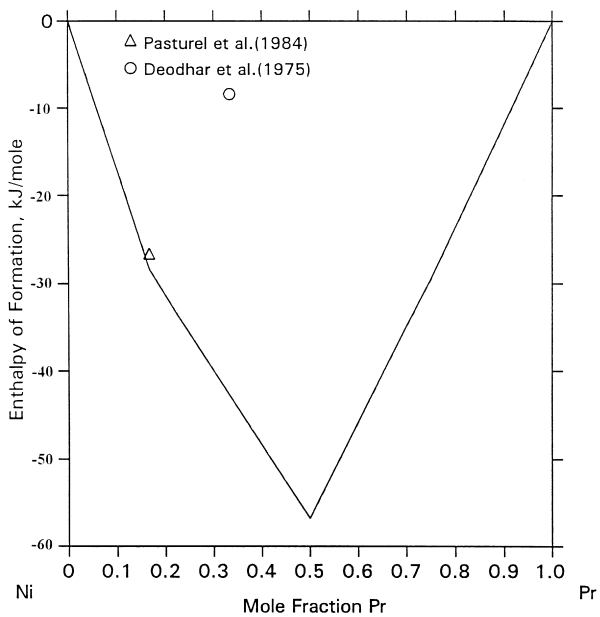

Fig. 6. Calculated enthalpy of formation in the Ni–Pr system with experimental measurements [15,35].

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